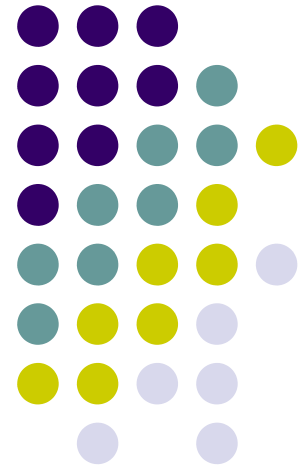
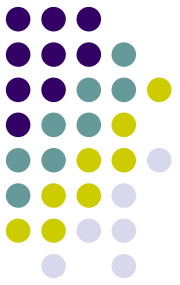


# Chapter 17

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## Temperature





# Outline for W134,D31

(Ch. 17) Temperature – relate to kinetic energy

Phases of matter

Zeroth law of thermodynamics

Temperature scales and conversions

Thermometers based on ...

Thermal expansion

## Homework

Ch. 17 Read 17.1-17.4, 17.6-17.9

P. 3,4,8,10,11,20,30,33,34,35,37,43

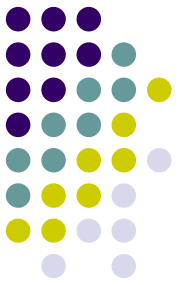
Ch. 18 Read 18.1 P. 1,6

Due next Mon

Notes: Exam 2 scores and review on Wednesday.

Labs: Thermal Expansion (see 17.4), Specific Heat (19.3)

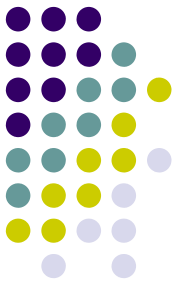
Special activity for Friday during my absence.



# Temperature

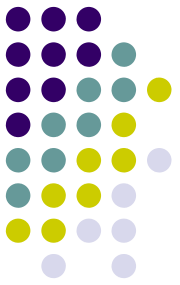
- 1) A measure of the internal energy (per mass) of an object.
  - 2) A measure of the average kinetic energy per particle stored in the random motions of the atoms and molecules in an object.
- Ex)  $\langle \frac{1}{2}mv^2 \rangle = \frac{3}{2} k_B T$  for monatomic ideal gases. (More on this in Ch. 18.)
  - How well can we judge temperatures through the sense of touch?
  - Is heat the same as internal energy? Ans: no  
 $\Delta E_{\text{int}} = Q = mc\Delta T$  (if no work,  $W$ )

# Temperature and phases of matter



- 1) Visualize the 3 main phases of matter on molecular scales (about  $10^{-9}$  m).
- 2) Solids
- 3) Liquids
- 4) Gases (or plasmas)

# Atomic Theory of matter (17-1)



Masses of atoms (atomic masses) are usually measured in *unified mass units*, u.

$$1 \text{ u} = 1.6605 \times 10^{-27} \text{ kg} = .001 \text{ kg} / 6.02 \times 10^{23} = 10^{-3} / N_A$$

where  $N_A$  = “Avagadro’s number” =  $6.02 \times 10^{23}$

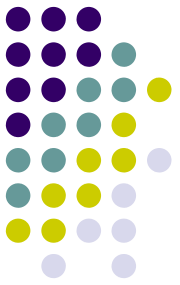
We define  $M$  = “molecular mass”, the mass of one particle, or the mass (in g) per mole of particles.

Also,  $m$  = total mass of a collection of particles.

Ex) Hydrogen has  $M=1.0079 \text{ u}$ , so  $N_A$  H atoms has 1.0079 grams of mass.

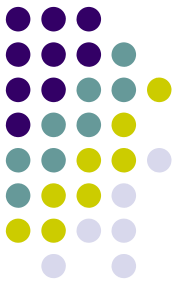
Ex) Carbon has  $M=12.0000 \text{ u}$ , so  $N_A$  C atoms has 12.0 g of mass.

Key relation between  $n, m$ , and  $M$ :  **$n \text{ (moles)} = m/M$**  (see 17-7)



# Thermal Contact

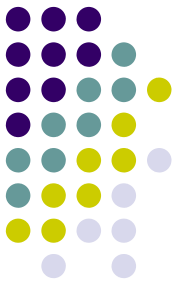
- Two objects are in **thermal contact** with each other if heat energy can be exchanged between them
  - **Heat transfer** can happen through conduction, radiation, or convection. Conduction requires direct contact. Convection requires a fluid.
  - A net energy is exchanged when there is a temperature difference between the two.



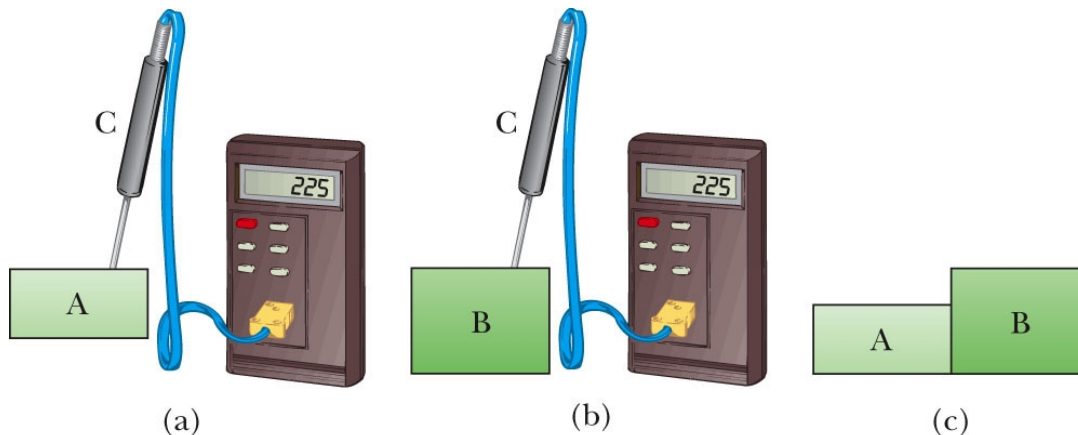
# Thermal Equilibrium

- **Thermal equilibrium** is the situation in which no heat flows between two objects when they are placed in thermal contact
  - The “thermal contact” allows conduction and/or radiation.

# Zeroth Law of Thermodynamics

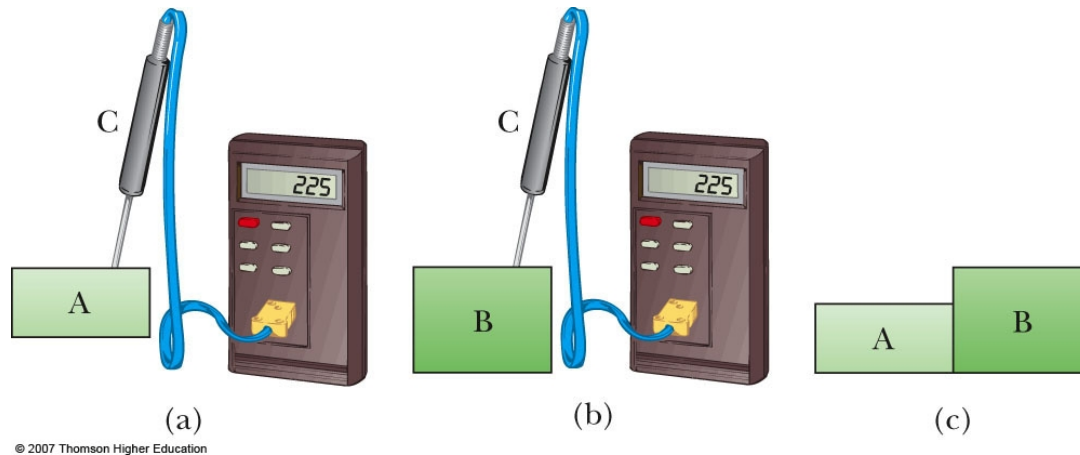
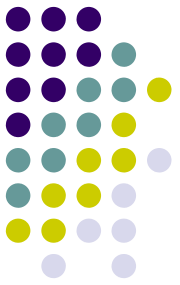


- If objects A and B are separately in thermal equilibrium with a third object C, then A and B are in thermal equilibrium with each other
  - Let object C be the thermometer
  - Since they are in thermal equilibrium with each other, there is no energy exchanged among them



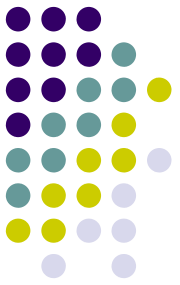


# Zeroth Law of Thermodynamics, Example



- Object C (thermometer) is placed in contact with A until they achieve thermal equilibrium
  - The reading on C is recorded
- Object C is then placed in contact with object B until they achieve thermal equilibrium
  - The reading on C is recorded again
- If the two readings are the same, A and B are also in thermal equilibrium

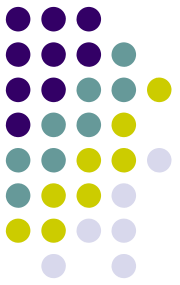
# Temperature – more definitions



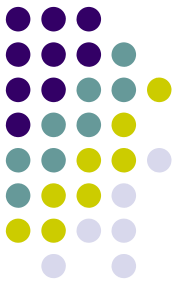
- **Temperature** can be thought of as the property that determines whether an object is in thermal equilibrium with other objects
- Two objects in thermal equilibrium with each other are at the same temperature.

# Temperature Scales

## Fahrenheit Scale

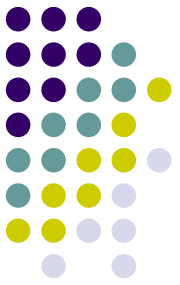


- A common scale in everyday use in the US
- Named for Daniel Fahrenheit
- Temperature of the ice point is  $32^{\circ}\text{F}$
- Temperature of the steam point is  $212^{\circ}\text{F}$
- There are 180 divisions (degrees) between the two reference points



# Celsius Scale

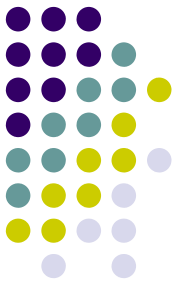
- The ice point of water is defined to be  $0^{\circ}\text{C}$
- The steam point of water is defined to be  $100^{\circ}\text{C}$
- The length of the column between these two points is divided into 100 increments, called degrees



# Absolute Temperature Scale

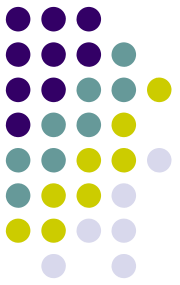
- Absolute zero is used as the basis of the absolute temperature scale, in Kelvin
- The size of the degree on the Kelvin scale is the same as the size of the degree on the Celsius scale
- To convert:
  - $T_C = T_K - 273.15$

# Absolute Temperature Scale, 2



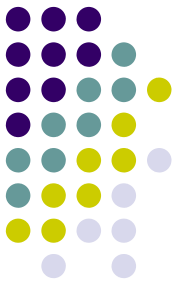
- The absolute temperature scale is now based on two new fixed points
  - Adopted by in 1954 by the International Committee on Weights and Measures
  - One point is absolute zero
  - The other point is the **triple point** of water
    - This is the combination of temperature and pressure where ice, water, and steam can all coexist

# Absolute Temperature Scale, 3



- The triple point of water occurs at  $0.01^{\circ}\text{C}$  and 4.58 mm of mercury
- This temperature was set to be 273.16 on the absolute temperature scale
  - This made the old absolute scale agree closely with the new one
  - The units of the absolute scale are **kelvins**

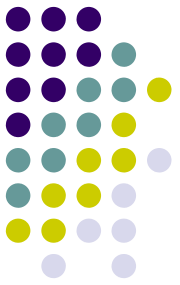
# Absolute Temperature Scale, 4



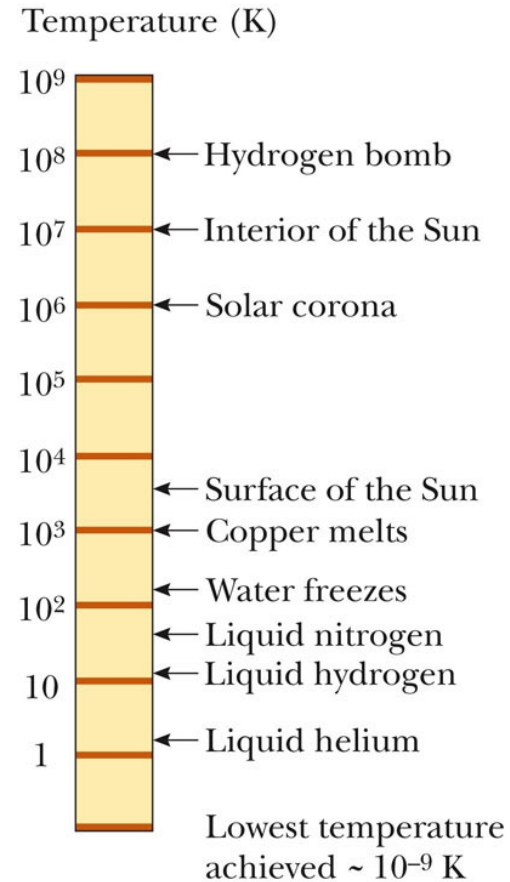
- The absolute scale is also called the Kelvin scale
  - Named for William Thomson, Lord Kelvin
- The triple point temperature is 273.16 K
  - No degree symbol is used with kelvins
- The kelvin is defined as  $1/273.16$  of the difference between absolute zero and the temperature of the triple point of water

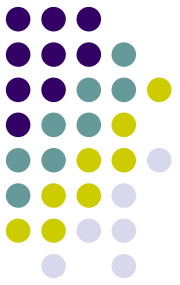


# Some Examples of Absolute Temperatures



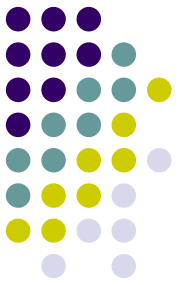
- The figure at right gives some absolute temperatures at which various physical processes occur
- The scale is logarithmic
- The temperature of absolute zero cannot be achieved
  - Experiments have come close





# Comparison of Scales

- Celsius and Kelvin have the same size degrees, but different starting points
  - $T_C = T - 273.15$
- Celsius and Fahrenheit have different sized degrees and different zero points



# Comparison of Scales, cont

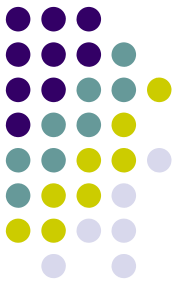
- Use Freezing and Boiling points of water to figure out conversion formulas

- Freezing point temperatures

$$0^{\circ}\text{C} = 273.15 \text{ K} = 32^{\circ} \text{ F}$$

- Steam point temperatures

$$100^{\circ}\text{C} = 373.15 \text{ K} = 212^{\circ} \text{ F}$$



# Outline for W14,D2

Temperature scales and conversions

Thermometers based on ...

Thermal expansion

Ideal Gas Law

Exam 2 review

## Homework

Ch. 17 Read 17.1-17.4, 17.6-17.9

P. 3,4,8,10,11,20,30,33,34,35,37,43

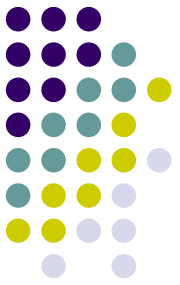
Ch. 18 Read 18.1 P. 1,6

Due next Mon

Notes: Exam 2 averages = 61.6%, 37.3%

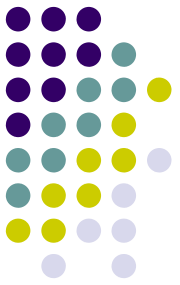
Labs: Thermal Expansion (see 17.4), Specific Heat (19.3)

Activity for Friday during my absence: follow the examples on slides 64-66 to solve similar problems.



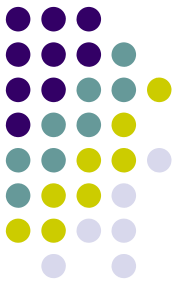
# Thermometers

- A **thermometer** is a device that is used to measure the temperature of a system
- Thermometers are based on the principle that some physical property of a system changes as the system's temperature changes.
- The physical property needs to change continuously, so we don't use phase changes to make thermometers.



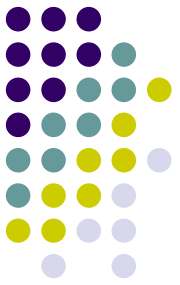
# Thermometers, cont

- These properties include:
  - The dimensions of a solid
  - The volume of a liquid
  - The electrical properties of conductors
  - The pressure of a gas at a constant volume
  - The color (glow) of an object
- A temperature scale can be established on the basis of any of these physical properties



# Thermal Expansion

- Thermal expansion is the increase in the size of an object with an increase in its temperature
- Thermal expansion is a consequence of the change in the average separation between the atoms in an object



# Linear Thermal Expansion

- Assume an object has an initial length  $L$
- That length increases by  $\Delta L$  as the temperature changes by  $\Delta T$
- The formula is  $\Delta L = \alpha L_i \Delta T$
- We define the **coefficient of linear expansion** as

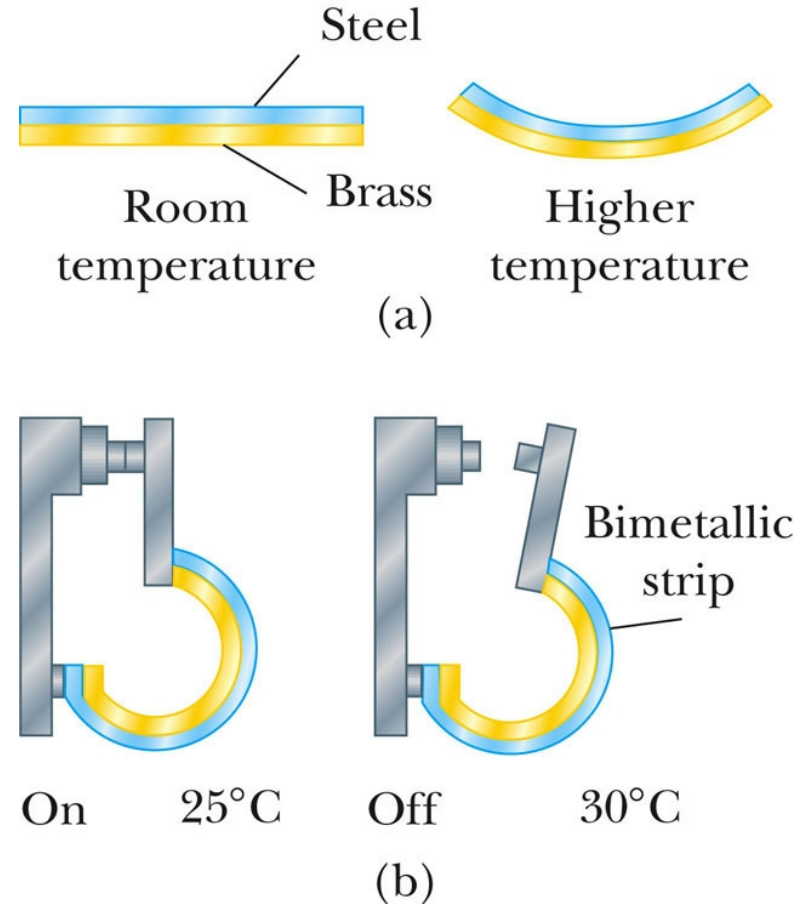
$$\alpha = \frac{\Delta L / L_i}{\Delta T}$$

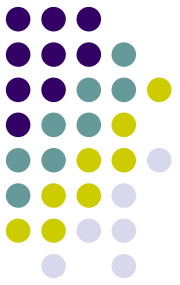
- This fractional change in length per change in temperature is a property of the *material*.



# Bimetallic Strip

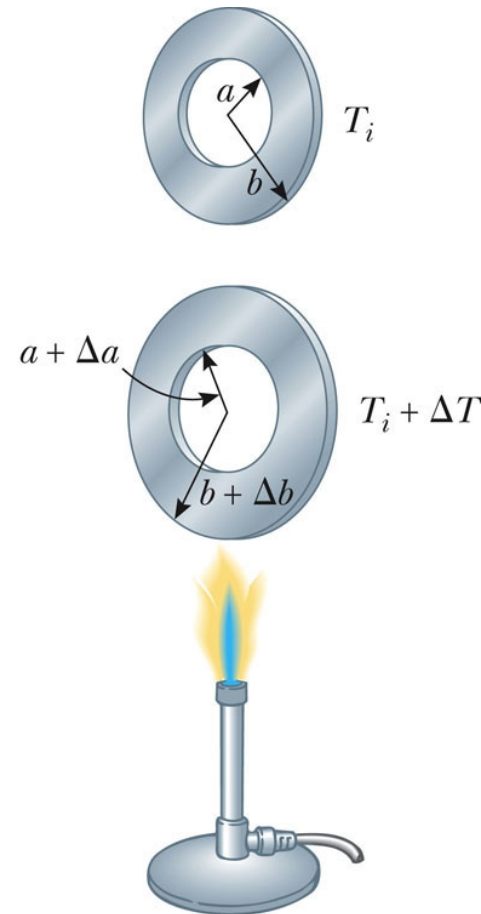
- Each substance has its own characteristic coefficient of expansion
- This can be made use of in the device shown, called a bimetallic strip
- DEMO
- It can be used in a thermostat

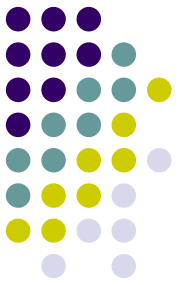




# Thermal Expansion, example

- As the washer shown at right is heated, all the dimensions will increase
- A cavity in a piece of material expands in the same way as if the cavity were filled with the material
- The expansion is exaggerated in this figure

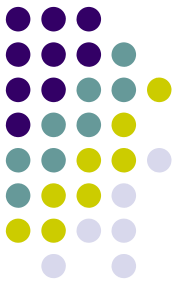




# Linear Expansion, cont

- The formula can also be written in terms of the initial and final conditions of the object:
  - $L_f - L_i = \alpha L_i (T_f - T_i)$  or ...
  - $L_f = L_i (1 + \alpha \Delta T)$
- The coefficient of linear thermal expansion,  $\alpha$ , has units of  $(^{\circ}\text{C})^{-1}$
- Here is a table of thermal expansion coefficients ...

# Some Coefficients

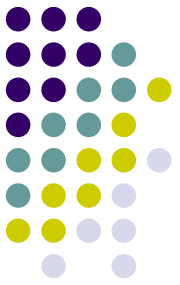


**TABLE 19.1**

**Average Expansion Coefficients for Some Materials Near Room Temperature**

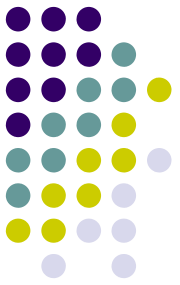
| Material            | Average Linear Expansion Coefficient ( $\alpha$ ) ( $^{\circ}\text{C}$ ) <sup>-1</sup> | Material                | Average Volume Expansion Coefficient ( $\beta$ ) ( $^{\circ}\text{C}$ ) <sup>-1</sup> |
|---------------------|----------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------|
| Aluminum            | $24 \times 10^{-6}$                                                                    | Alcohol, ethyl          | $1.12 \times 10^{-4}$                                                                 |
| Brass and bronze    | $19 \times 10^{-6}$                                                                    | Benzene                 | $1.24 \times 10^{-4}$                                                                 |
| Copper              | $17 \times 10^{-6}$                                                                    | Acetone                 | $1.5 \times 10^{-4}$                                                                  |
| Glass (ordinary)    | $9 \times 10^{-6}$                                                                     | Glycerin                | $4.85 \times 10^{-4}$                                                                 |
| Glass (Pyrex)       | $3.2 \times 10^{-6}$                                                                   | Mercury                 | $1.82 \times 10^{-4}$                                                                 |
| Lead                | $29 \times 10^{-6}$                                                                    | Turpentine              | $9.0 \times 10^{-4}$                                                                  |
| Steel               | $11 \times 10^{-6}$                                                                    | Gasoline                | $9.6 \times 10^{-4}$                                                                  |
| Invar (Ni–Fe alloy) | $0.9 \times 10^{-6}$                                                                   | Air <sup>a</sup> at 0°C | $3.67 \times 10^{-3}$                                                                 |
| Concrete            | $12 \times 10^{-6}$                                                                    | Helium <sup>a</sup>     | $3.665 \times 10^{-3}$                                                                |

<sup>a</sup> Gases do not have a specific value for the volume expansion coefficient because the amount of expansion depends on the type of process through which the gas is taken. The values given here assume the gas undergoes an expansion at constant pressure.



# Area Expansion

- The change in area is proportional to the original area and to the change in temperature:
  - $\Delta A = 2\alpha A_i \Delta T$
  - Shouldn't there be an  $\alpha^2$  in this equation?



# Volume Expansion

- The change in volume is proportional to the original volume and to the change in temperature

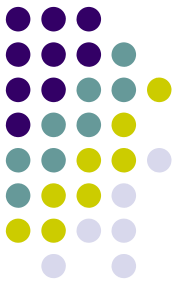
$$\Delta V = \beta V_i \Delta T$$

$\beta$  is the coefficient of volume expansion

For a solid,  $\beta=3\alpha$

This assumes the material is isotropic, expanding the same in all directions.

For a liquid or gas,  $\beta$  is given in the table



# Thermometer, Liquid in Glass

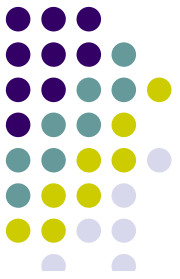
- A common type of thermometer is a liquid-in-glass
- The material in the capillary tube expands as it is heated
- The liquid is usually mercury or alcohol



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# Electrical properties that depend on T.



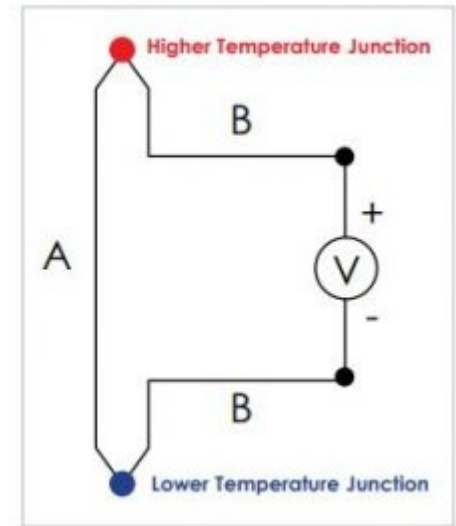
Resistance increases with temperature for typical metal conductors (Iron, silver, copper, etc) and semiconductors. This can form the basis of a thermometer or *thermistor*. Here's the temperature dependence of resistivity,  $\rho$ :

$$\rho_T = \rho_0 [1 + \epsilon(T - T_0)]$$

## The Seebeck Effect and Thermocouple Thermometers

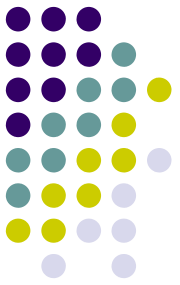
A thermocouple is formed by the junction of two dissimilar metals (e.g., Fe for A, and Cu for B). They are self-powered because the thermocouple creates a small thermoelectric voltage known as the Seebeck Effect. This voltage is proportional to the difference in the temperature between the hot end and the cold end of the thermocouple. As a result, the instrument requires a known temperature at one end of the thermocouple to calculate the temperature at the other end.

The Seebeck Effect

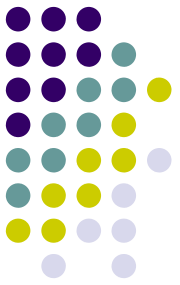




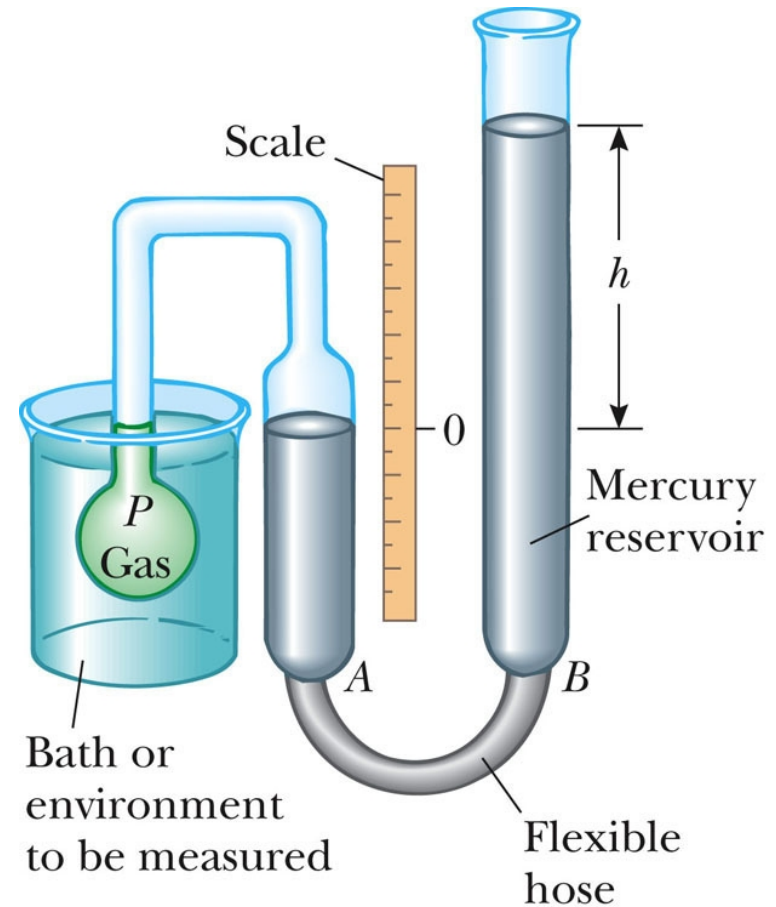
# Thermal Radiation



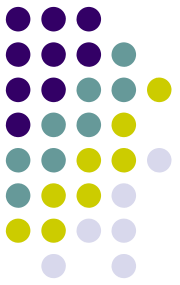
# Gas Pressure increases with T: The Constant-Volume Gas Thermometer



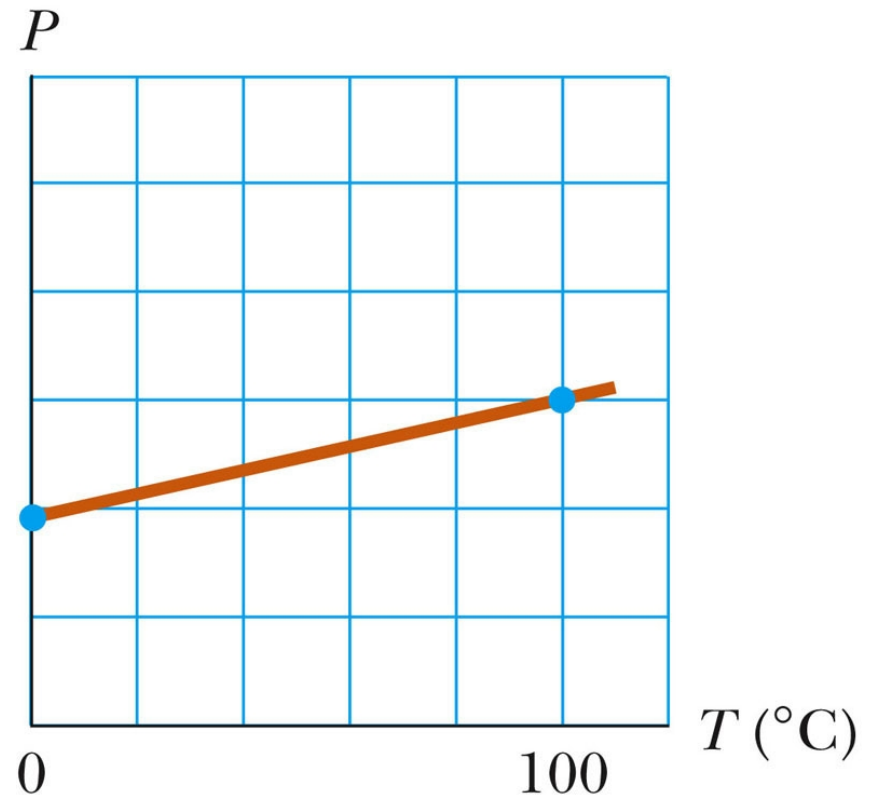
- The physical change exploited is the variation of pressure of a fixed volume gas as its temperature changes
- The volume of the gas is kept constant by raising or lowering the reservoir B to keep the mercury level at A constant



# Constant-Volume Gas Thermometer, DEMO

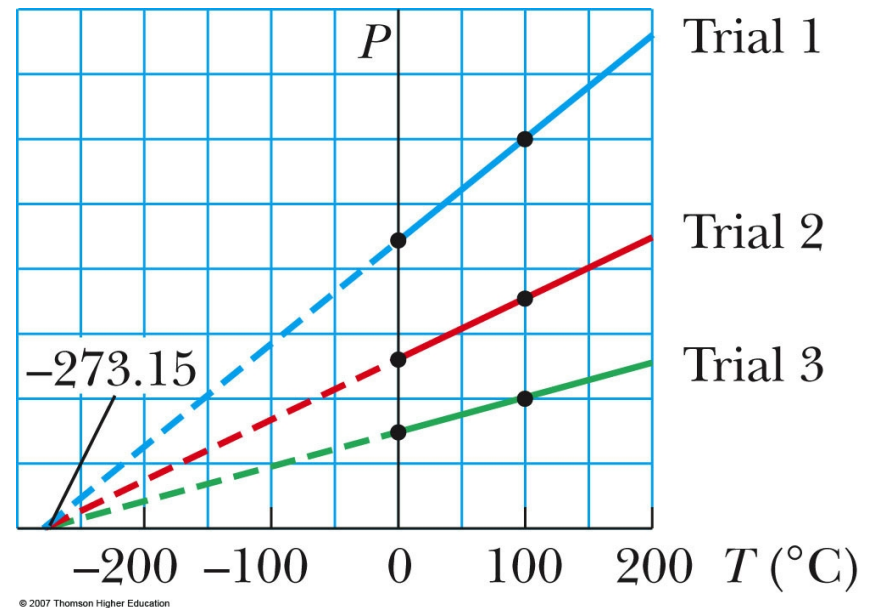


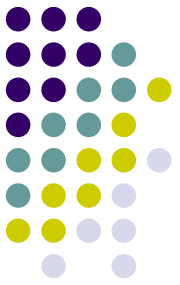
- To find the temperature of a substance, the gas flask is placed in thermal contact with the substance
- The pressure is found on the graph
- The temperature is read from the graph



# Absolute Zero

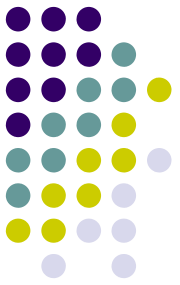
- The thermometer readings are virtually independent of the gas used
- If the lines for various gases are extended, the pressure is always zero when the temperature is  $-273.15^{\circ}\text{C}$
- This temperature is called **absolute zero**





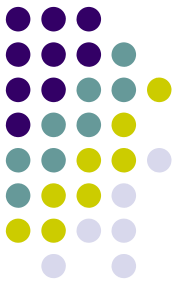
# An Ideal Gas

- Has no interatomic forces acting between particles. (In reality, these are weak unless gas is ionized.)
- Has elastic collisions.
- Particles are tiny compared to their separations (so the density is low).
- Obey's the “Ideal Gas Law”,  $PV=nRT$
- Air is approximately “ideal”.



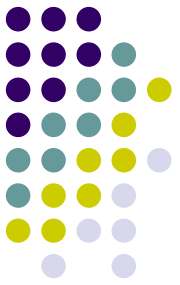
# Gas: Equation of State

- It is useful to know how the volume, pressure and temperature of the gas are related
- An equation that interrelates these quantities is called an *equation of state*
  - If the gas is maintained at a low density, the equation of state becomes the **Ideal Gas Law**:  
$$PV = nRT$$
- Other versions ...  $PV = NK_B T$ ,  $P = \rho K_B T$



# The Mole

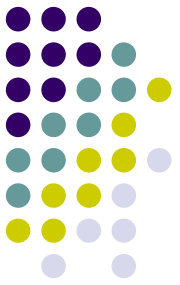
- The amount of gas in a given volume is conveniently expressed in terms of the number of moles
- One **mole** of any substance is that amount of the substance that contains **Avogadro's number** of constituent particles
  - Avogadro's number  $N_A = 6.022 \times 10^{23}$
  - The constituent particles can be atoms or molecules



# Moles, cont

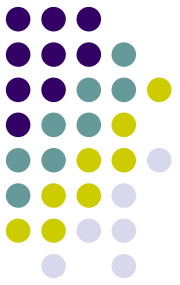
- The number of moles can be determined from the mass,  $m$ , of the gas:  $n = m / M$ 
  - $M$  is the molar mass of the substance
    - $M$  can be obtained from the periodic table
    - $M$  is the atomic mass expressed in grams/mole
      - Example: He gas has mass of 4.00 u so  $M = 4.00$  g/mol or .004 kg/mol.
  - $m$  is the mass of the sample
  - $n$  is the number of moles
  - Example: How much mass does 3 mol of  $N_2$  have?
    - Sol:  $m = nM$ .  $N_2$  has mass of 28.0 u so  $M = 28.0$  g/mol.  
Thus,  $m = 3(28\text{g}) = 84$  g.





# Gas Laws (derived from $PV=nRT$ )

- When a gas is kept at a constant temperature, its pressure is inversely proportional to its volume (Boyle's law)
- When a gas is kept at a constant pressure, its volume is directly proportional to its temperature (Charles and Gay-Lussac's law)
- When the volume of the gas is kept constant, the pressure is directly proportional to the temperature (Gay-Lussac's law)



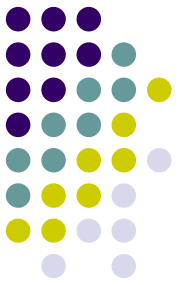
# Ideal Gas Law

- The equation of state for an ideal gas combines and summarizes the other gas laws

$$PV = nRT$$

- This is known as the **ideal gas law**
- $R$  is a constant, called the Universal Gas Constant
  - $R = 8.314 \text{ J/mol}\cdot\text{K} = 0.08214 \text{ L} \cdot \text{atm/mol}\cdot\text{K}$
- From this, you can determine that 1 mole of any gas at atmospheric pressure and at 0°C is 22.4 L.

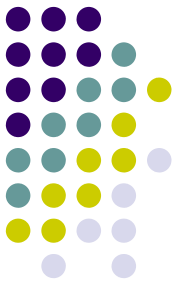
$$\text{Show: } V = nRT/P = (1\text{mol})(0.08214)(273\text{K})/(1\text{atm}) \\ = 22.4 \text{ L}$$



# Ideal Gas Law, cont

- The ideal gas law is often expressed in terms of the total number of molecules,  $N$ , present in the sample
- $PV = nRT = (N/N_A) RT = Nk_B T$ 
  - $k_B$  is Boltzmann's constant
  - $k_B = 1.38 \times 10^{-23} \text{ J/K}$
- It is common to call  $P$ ,  $V$ , and  $T$  the **thermodynamic variables** of an ideal gas

# Ideal Gas Law - Examples



Ex) P. 34. In an internal combustion engine, air at atmospheric pressure and temperature of about 20°C is compressed in the cylinder by a piston to 1/9 of its original volume. Estimate the temperature of the compressed air, assuming the pressure reaches 40 atm.

$$\text{Soln: } P_i V_i / T_i = P_f V_f / T_f \rightarrow T_f = (P_f V_f / P_i V_i) T_i = (40 * 1/9) T_i$$

$$\text{Use } T_i = 273 + 20 = 293 \text{ K. } T_f = (40/9) * 293 = 1302 \text{ K}$$

# Ideal Gas Law - Examples



Ex) P. 37. Calculate the density of nitrogen at STP using the ideal gas law.

Soln: Need to know “nitrogen” means  $N_2$  and that  $N_2$  has molar mass of  $M=28$  g/mol or .028 kg/mol.

Density,  $\rho = m/V$

STP  $\rightarrow T=273$  K,  $P=101,000$  Pa

$PV = nRT$

Take  $n=1$  mol, then  $m=.028$  kg. Just need  $V$  ...

$V = nRT/P = 1(8.314)(273)/101000 = .02246$  m<sup>3</sup> (true for all gases).

So  $\rho = m/V = .028/.02246 = 1.25$  kg/m<sup>3</sup>

# Ideal Gas Law - Examples



Ex) P. 43. A sealed metal container contains a gas at 30°C and absolute pressure of 1.0 atm. To what temperature must the gas be heated for the pressure to double to 2.0 atmospheres? (Ignore the expansion of the container.)

Soln:  $V$  fixed  $\rightarrow$  Gay-Lussac Law,  $P \propto T$ .

Convert  $T_i = 30 + 273 = 303$  K.

Use ratios.

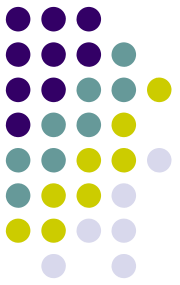
$$P_i/P_f = T_i/T_f$$

$$1/2 = 303/T_f$$

$$T_f = 303/0.5 = 606 \text{ K}$$

$$T_f = 606 - 273 = 333^\circ\text{C}$$

# Kinetic Theory (Ch. 18)



See textbook's derivation of this expression for pressure:

$$P = \frac{1}{3} \frac{Nm \overline{v^2}}{V}$$

$N$  = number of molecules in container

$m$  = mass of the gas molecules

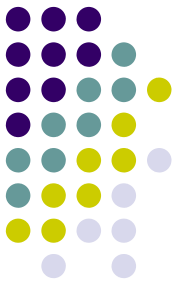
$\overline{v^2}$  = mean speed squared. Note, the  $v$  is called  $v_{\text{RMS}}$

$V$  = volume of container

If all of the molecules have the same mass, then the average translational kinetic energy is given by

$$\text{KE}_{\text{trans}} = \overline{K} = \left\langle \frac{1}{2} m v^2 \right\rangle = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$

# Kinetic Theory (Ch. 18)



Ex) Express the pressure in terms of the average translational kinetic energy.

$$P = \frac{1}{3} \frac{Nm \overline{v^2}}{V}$$
$$P = \frac{2}{3} \frac{N \left( \frac{1}{2} \right) m \overline{v^2}}{V}$$
$$P = \frac{2}{3} N \frac{\overline{K}}{V}$$

P. 18.1) Calculate the rms speed of He atoms near the surface of the Sun at a temperature of about 6000 K.

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} kT \quad v_{\text{RMS}} = (3 kT/m)^{1/2}$$

Need  $M_{\text{He}} = 4 \text{ u}$ , and  $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$

$$v_{\text{RMS}} = [(3 \cdot 1.38 \times 10^{-23} \cdot 6000) / (6.64 \times 10^{-27})]^{1/2} = 6100 \text{ m/s}$$



# Kinetic Theory (Ch. 18)



Ex) P. 18.6 (a) What is the average translational kinetic energy of a nitrogen molecule at STP?

$$\left\langle \frac{1}{2} m v^2 \right\rangle = \frac{3}{2} kT \quad K_{\text{trans}} = \frac{3}{2} (1.38 \times 10^{-23})(273) = 5.65 \times 10^{-21} \text{ J}$$

(b) What is the total translational kinetic energy of 1.0 mol of  $\text{N}_2$  molecules at  $25^\circ\text{C}$ ?

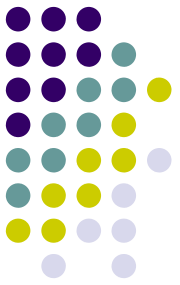
$$N \frac{1}{2} m \overline{v^2} = N \frac{3}{2} kT \quad K_{\text{tot,trans}} = 6.02 \times 10^{23} \cdot \frac{3}{2} (1.38 \times 10^{-23})(298\text{K}) = 3713 \text{ J}$$

(myQ) What is the average translational kinetic energy of ANY other ideal gas at STP?

(myQ) What is the speed ratio of  $\text{H}_2$  to He in a mixture at STP?

$$m_{\text{H}} v_{\text{H}}^2 = m_{\text{He}} v_{\text{He}}^2 \quad \text{so } v_{\text{H}}/v_{\text{He}} = (m_{\text{He}}/m_{\text{H}})^{1/2} = 2^{1/2}$$

# First Law of Thermodynamics (Ch. 19)



1<sup>st</sup> Law is just Conservation of Energy applied to heat.

- Recall:  $\Delta K + \Delta U = -f_k d + W_{\text{ext}} + Q + T + T + T + T \dots$
- Replace  $f_k d = \Delta E_{\text{int}}$ . In gas systems,  $\Delta K + \Delta U = 0$ .
- In Giancoli:  $W$  = work done **by** the gas, so  $W_{\text{ext}} = -W$
- Then you're left with:  **$\Delta E_{\text{int}} = Q - W$**
- For a closed system,  $\Delta E_{\text{int}} = 0$ .
- $Q$  = heat added to the system,  $W$  = work done **by** the system.

# First Law of Thermodynamics (Ch. 19)



$$\Delta E_{\text{int}} = Q - W \quad (\text{cont.})$$

Ways to evaluate the Q...

1)  $Q = mc\Delta T$  (e.g., in *calorimetry*)

- $m$  = mass of object
- $\Delta T = T_f - T_i$
- $c$  = specific heat capacity = amt of energy required to raise the temperature of 1 g (or 1 kg) of substance by 1°C (or 1K).

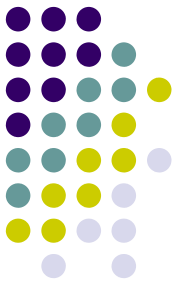
$$\text{Ex) } c_{\text{H}_2\text{O, liquid}} = 1.000 \text{ cal/g}^\circ\text{C} \quad \text{or } 4186 \text{ J/kgK}$$

$$\text{Ex) } c_{\text{steel}} = 0.110 \text{ cal/g}^\circ\text{C} \quad \text{or } 452 \text{ J/kgK}$$

2)  $Q = \pm mL_f$  (phase changes of melting and freezing)

- $L_f$  = latent heat of fusion
- Use + for melting, use – for freezing.

# First Law of Thermodynamics (Ch. 19)



$$\Delta E_{\text{int}} = Q - W \quad (\text{cont.})$$

Ways to evaluate the Q...

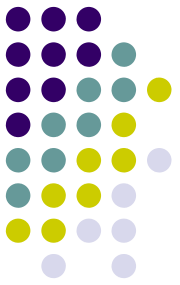
3)  $Q = \pm mL_v$  (phase changes of boiling and condensing)

- $L_v$  = latent heat of vaporization
- Use + for boiling, use – for condensing.

Ex) How much heat is required to warm up and melt a 2kg block of ice starting at  $T_i = -20^\circ\text{C}$  and ending at  $T_f = 0^\circ\text{C}$  in liquid form.

$$\begin{aligned} \text{Ans: } Q_{\text{tot}} &= Q_{\text{warm}} + Q_{\text{melt}} \\ &= m_{\text{ice}} c_{\text{ice}} (0 - -20) + L_f m \\ &= 2\text{kg}(2100\text{J/kgC})(20) + (333,000 \text{ J/kg})(2\text{kg}) \\ &= 84000 + 666000 = 750 \text{ kJ} \end{aligned}$$

# First Law of Thermodynamics (Ch. 19)



$$\Delta E_{\text{int}} = Q - W \quad (\text{cont.})$$

Finding the specific heat of a substance (e.g., copper) using calorimetry. (See Example 19-5)

Hot copper is added to water + Al cup in a calorimeter.

Assume calorimeter is perfectly insulated, then:

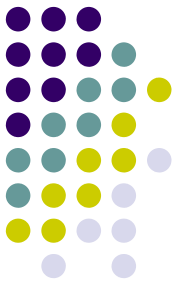
Heat lost by copper = heat gained by (water + Al)

$$-Q_{\text{Cu}} = Q_{\text{w}} + Q_{\text{Al}}$$

$$-m_{\text{Cu}} c_{\text{Cu}} \Delta T_{\text{Cu}} = m_{\text{w}} c_{\text{w}} \Delta T + m_{\text{Al}} c_{\text{Al}} \Delta T$$

$$c_{\text{Cu}} = \frac{m_{\text{w}} c_{\text{w}} \Delta T + m_{\text{Al}} c_{\text{Al}} \Delta T}{-m_{\text{Cu}} \Delta T_{\text{Cu}}}$$

# First Law of Thermodynamics (Ch. 19)



Applying  $\Delta E_{\text{int}} = Q - W$  to gases.

- $Q = mc\Delta T$  does not apply to gas systems!
- Gases can expand and do work on the walls of the container.
- Recall  $W$  = work done BY the gas.
- There are only two easy cases where  $Q$  relates to  $\Delta T$ :
  - 1)  $Q_V = nC_V\Delta T$  when  $V$  is fixed  
here  $C_V$  = molar heat capacity @ constant  $V$ , in J/mol•K
  - 2)  $Q_P = nC_P\Delta T$  when  $P$  is fixed  
here  $C_P$  = molar heat capacity @ constant  $P$ , in J/mol•K
- Otherwise
  - 3)  $Q = \Delta E_{\text{int}} + W$  (always true)
  - 4)  $\Delta E_{\text{int}} = nC_V\Delta T$  (always true)

# First Law of Thermodynamics (Ch. 19)



Applying  $\Delta E_{\text{int}} = Q - W$  to gases.

Ex) How much heat must be added to the air in a constant volume gas thermometer if the bulb contains 0.05 mol of air, and it changes from 0 to 100° C?

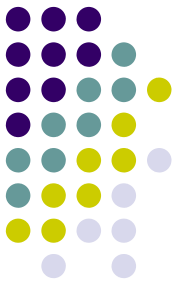
Isovolumetric case

Soln: since  $V$  is constant,  $Q = nC_v\Delta T$  applies.

Need  $C_v$  for air...  $C_v = 4.97 \text{ cal/mol}\cdot\text{K}$  (4.96 for  $\text{N}_2$ )

$Q = (0.05 \text{ mol})(4.97)(100\text{K}) = 24.85 \text{ cal}$  or 104.0 J

# First Law of Thermodynamics (Ch. 19)



Applying  $\Delta E_{\text{int}} = Q - W$  to gases.

Dealing with volume changes, and work by the gas.

Work by a gas: 
$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} P dV$$

Ex) (Simplest case) How much work is done by a gas on a piston if it's volume increases from 1 to 3 L while the gas pressure remains fixed at 2 atm?

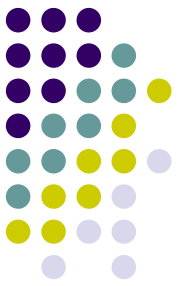
Isobaric case

Soln: Since  $P$  is constant,  $W = P \int dV = P \Delta V = 202,000 (V_f - V_i)$   
 $\Delta V = 3 \text{ L} - 1 \text{ L} = 2 \text{ L} \rightarrow .002 \text{ m}^3$ .

So  $W = 202,000 \cdot 0.002 = 404 \text{ J}$



# First Law of Thermodynamics (Ch. 19)



## PV diagrams and gas processes

Plot of pressure vs volume.

Assume  $n$  is fixed

Area under curve is work,  $|W|$

$$W = \int_{V_i}^{V_f} P dV$$

If process moves right  $\rightarrow$  expansion,  $W > 0$

If process moves left  $\rightarrow$  compression,  $W < 0$

Special cases: (see interactive SWF)

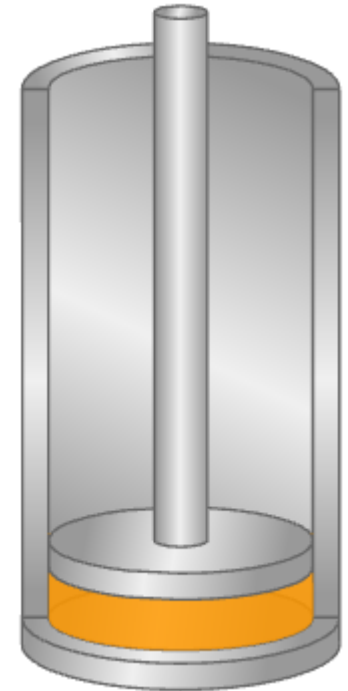
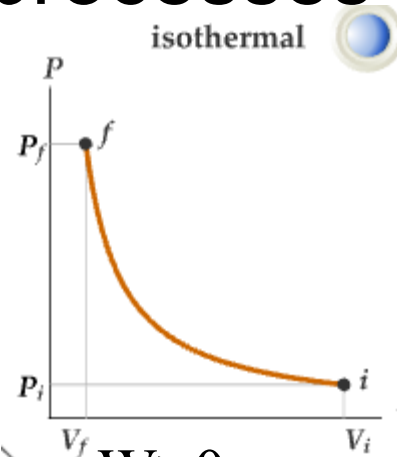
1) **Isobaric:** constant pressure.

$$W = P\Delta V$$

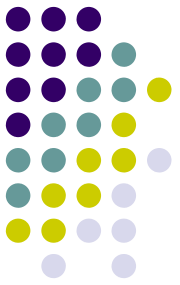
$$Q = nC_p\Delta T$$

$$\Delta E_{\text{int}} = Q - W \text{ can be written } \Delta E_{\text{int}} = nC_p\Delta T - P\Delta V$$

$$\text{Or } nC_v\Delta T = nC_p\Delta T - nR\Delta T$$



# First Law of Thermodynamics (Ch. 19)



## PV diagrams and gas processes

Special cases: (cont.)

2) **Isovolumetric:** constant volume.

$$W=0$$

$$Q=nC_V\Delta T$$

$$\Delta E_{\text{int}} = Q = nC_V\Delta T$$

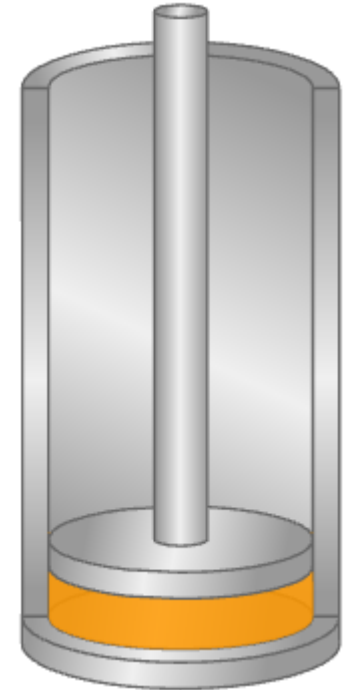
3) **Isothermal:** constant temperature.

$$P = nRT/V \quad (P \propto 1/V)$$

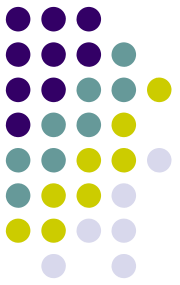
$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} \frac{nRT}{V} dV = nRT \ln \left( \frac{V_f}{V_i} \right)$$

$$\Delta E_{\text{int}} = Q - W = 0 \quad (\text{because } \Delta E_{\text{int}} = nC_V\Delta T \text{ for all processes})$$

$$\text{So } Q = W$$



# First Law of Thermodynamics (Ch. 19)



## PV diagrams and gas processes

Special cases: (cont.)

4) **Adiabatic:** no heat added.

$$Q=0$$

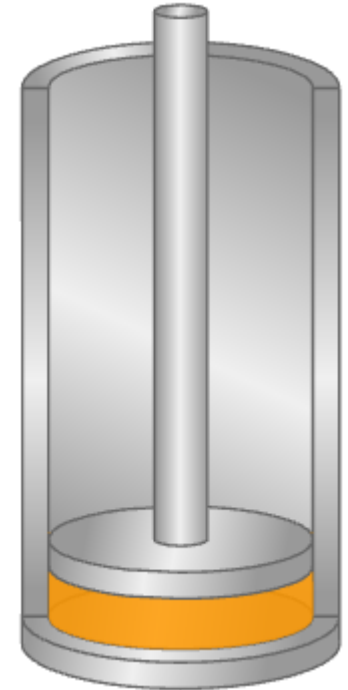
$$\Delta E_{\text{int}} = -W$$

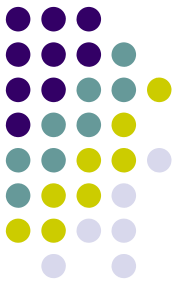
$$PV^\gamma = \text{const} \quad \text{and} \quad TV^{\gamma-1} = \text{const}$$

where  $\gamma = C_p/C_v$  (e.g.,  $\gamma=5/3$  monatomic)

To figure out  $W$ , invent constant  $C$  such that  $T=CV^{1-\gamma}$

$$W = \int_{V_i}^{V_f} P dV = \int_{V_i}^{V_f} \frac{nRT}{V} dV = \int_{V_i}^{V_f} \frac{nRCV^{1-\gamma}}{V} dV = \int_{V_i}^{V_f} nRCV^{-\gamma} dV$$





# Outline for W15,D3 – *Last Day!*

Final Exam Information

Examples of gas processes and cyclic processes

Review – questions on exam-like problems?

## Homework

Ch. 19 P. 1,6,8,9,20,21,32,33,35,36,40,49 Do before exam

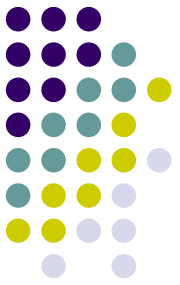
Notes:

“NEW STUFF” has: exam-like problems on Ch 19.

“NEW STUFF” has: homework keys for Chs. 17-19.

Course evaluations online.

Location of final exam is Science Annex 105 (downstairs).



# Final Exam Information

## Logistics

Time: Thursday, May 15, 4:15-6:15

Accommodations: email me if you haven't already

Place: SA 105

**Format** – like previous exams

Multiple choice

Expect 6-8 pages (a little longer than previous)

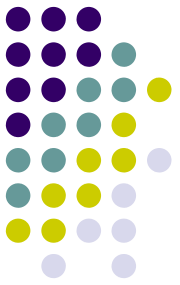
Last page has list of equations and constants.

Scratch paper available

Scantrons – bring No. 2 pencils, bring ID card

Bring calculator

No other aids.



# Final Exam Information

## Coverage

Comprehensive

~1/3 Exam I material Ch. 1-5 Units, Kinematics & Forces

~1/3 Exam II material Ch. 7-11 Work and energy, rotation

~1/3 Thermodynamics Ch. 17-19

See old notes on which sections are emitted from Chs 1-5,7-11.

For Chs. 17-19:

Omit 17.5, 17.10

Omit 18.2-18.8 (only read 18.1)

Omit 19.10

## Study resources

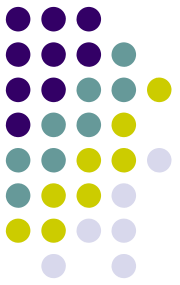
Exam I and Exam II - the bare minimum for the old material

“NEW STUFF”: Exam-like questions,

“Final Exam Review”, Equations as they will appear,

Practice quizzes, Hwk keys. PDF powerpoints

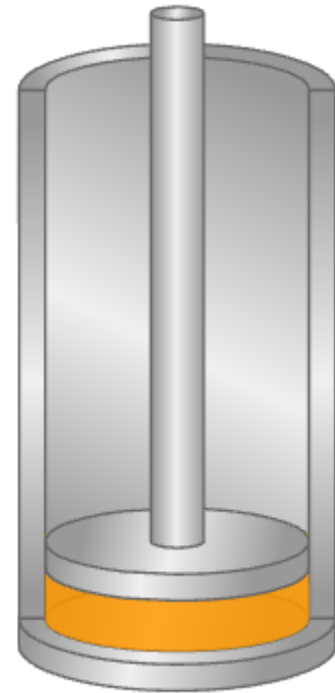
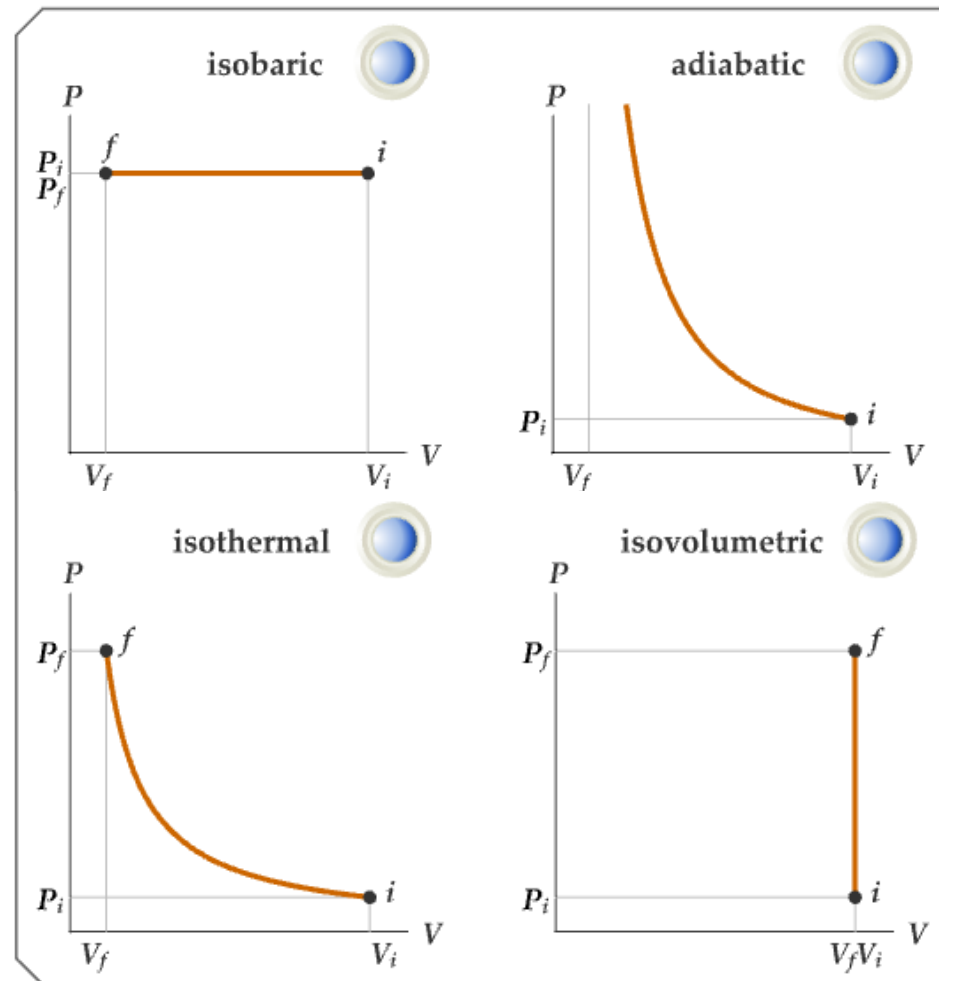
# First Law of Thermodynamics (Ch. 19)



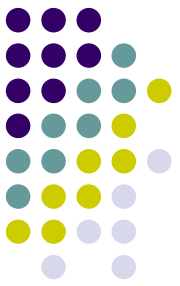
## PV diagrams and gas processes

### Special cases:

- 1) Isobaric
- 2) Isovolumetric
- 3) Isothermal
- 4) Adiabatic
- 5) Cyclic processes



# First Law of Thermodynamics (Ch. 19)



## PV diagrams and gas processes

### Special case: Cyclic processes (cont.)

A process that begins and ends in the same state is cyclic.

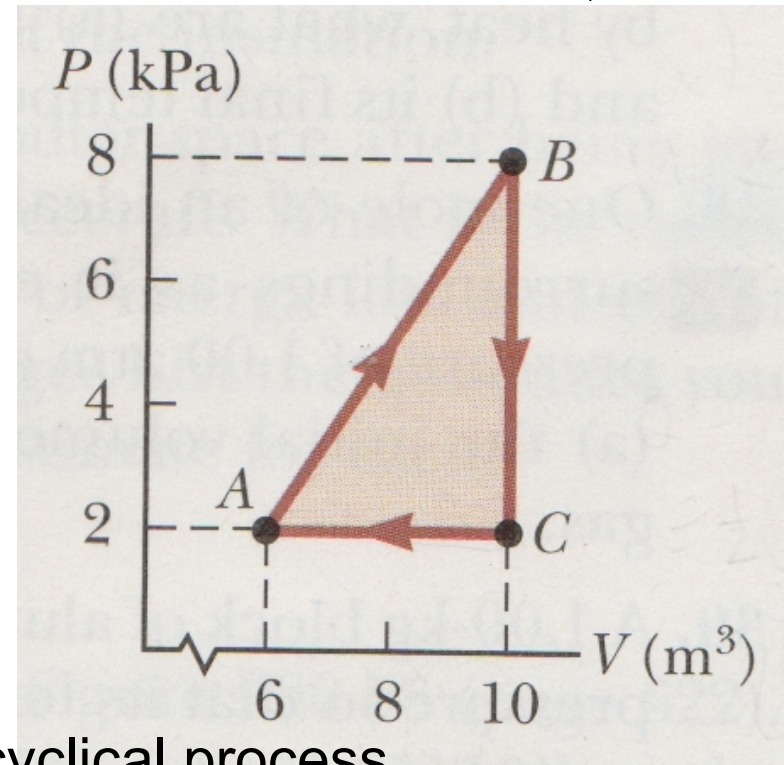
$$(P_i, V_i, T_i, n_i) = (P_f, V_f, T_f, n_f)$$

Since  $\Delta E_{\text{int}} = nC_V\Delta T$ ,  $\Delta E_{\text{int}} = 0$ , and  $W = Q$ .

Area inside loop =  $|W_{\text{net}}| = |Q_{\text{net}}|$ .

Clockwise  $\rightarrow W_{\text{net}} > 0$ .

Counterclockwise  $\rightarrow W_{\text{net}} < 0$ .



Example: The figure shows a 3-stage cyclical process.

Q: What is  $W$  from B to C?

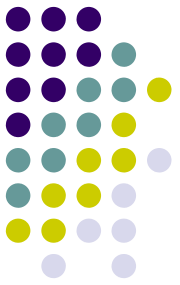
Q: What is  $W$  from C to A? Q: How does  $W_{AB}$  compare to  $W_{CA}$ ?

Q: Why is  $W_{\text{net}} > 0$  for a CW cycle?

Q: Which of the 3 sub-processes have names?



# First Law of Thermodynamics (Ch. 19)



## PV diagrams and gas processes

Special case: Cyclic processes (cont.)

Example: (cont.)

Q: What is  $W_{\text{net}}$ ?

$$A = \frac{1}{2} (6000 \text{ Pa})(4 \text{ m}^3)$$

$$\text{So } W = 12,000 \text{ J}$$

Q: What is  $Q_{\text{net}}$ ?

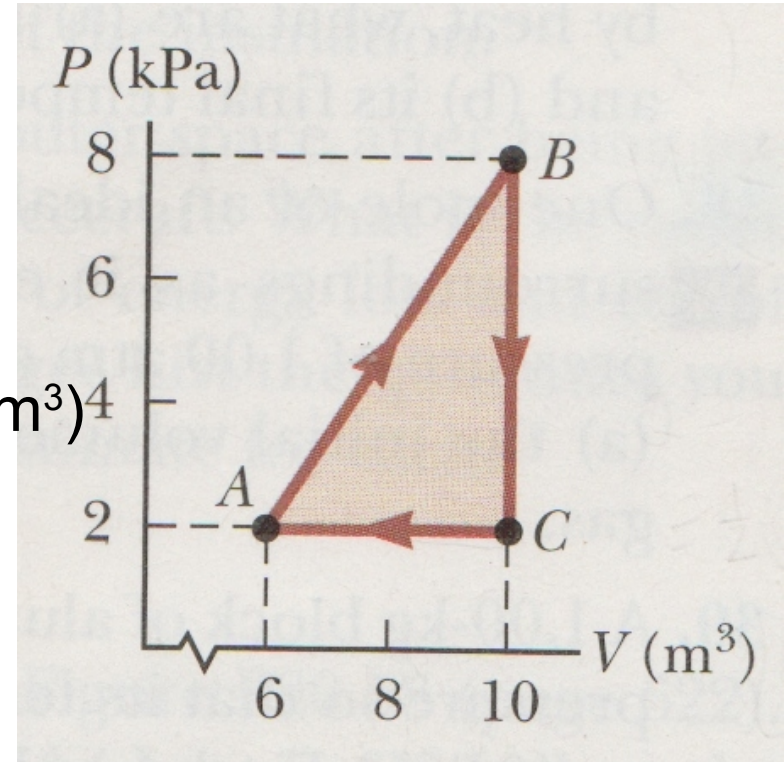
$$Q = W$$

Q: What is  $\Delta E_{\text{int}}$ ?

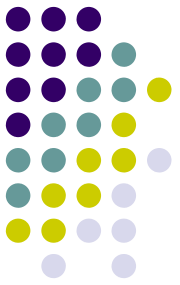
$$\text{So } Q = 12,000 \text{ J}$$

$$\Delta E_{\text{int}} = Q - W$$

$$\text{So } \Delta E_{\text{int}} = 0$$



# First Law of Thermodynamics (Ch. 19)



## Equipartition of energy and degrees of freedom.

Example: P. 49) What is the internal energy of 5.4 mol of an ideal diatomic gas at 2450 K, assuming all possible degrees of freedom are active?

Soln: Look at Fig. 19-17. It shows  $C_V$  for  $H_2$  as a function of temperature. At 2450 K, KE is stored equally in Translational (3), Rotational (2) and Vibrational (2) degrees of freedom. So  $C_V = (7 \times 1/2) R = 7/2 R$ .

So we can use  $E_{\text{int}} = nC_V T = n(7/2 R) T$   
 $= 5.4(3.5 \cdot 8.31) 2450 = 385,000 \text{ J}$

For a monatomic gas,  $C_V = 3/2 R$  regardless of temperature.

# First Law of Thermodynamics (Ch. 19)

## Exam-like question Review

